

Krishna Lal

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SUMMARY

AI/ML Engineer focused on machine learning systems, generative AI, LLMs, and AI research. Hands-on experience building and training ML models, AI agents, vector-based systems, and production-oriented AI backends using PyTorch, TensorFlow, Scikit-learn, FastAPI, and modern LLM frameworks. Experienced in experimentation, data processing, model evaluation, and open-source development.

EDUCATION

Haldia Institute of Technology — B.Tech, CSE - Data Science

2024 – Present | Haldia

CGPA: 8.7 (up to 5th Semester) | Graduation: 2028

PROJECTS

Vector – From-Scratch GPT-Style Language Model & Voice AI System

[Vector-AI](#)

- Built and trained a ~4.8M-parameter decoder-only transformer (6 layers, 8 heads, causal self-attention) from scratch in PyTorch with a custom tokenizer and no pretrained weights; reached validation perplexity of ~2.19.
- Designed a hybrid neural + symbolic pipeline with a deterministic arithmetic tool (100% accuracy vs 13.6% for the raw model), an evaluation script, and a voice interface (speech-to-text to model to text-to-speech).

Tools: Python, PyTorch, Transformers (from scratch), Custom Tokenizer, SpeechRecognition, pyttsx3

Real-Time CCTV Object Detection & Event Pipeline

[OpenCV Yolo](#)

- Built a video analytics pipeline with YOLO (Ultralytics) and OpenCV for object detection and multi-object tracking on CCTV-style footage, with confidence filtering and live FPS monitoring.
- Implemented restricted-zone entry detection that generates structured JSON event reports, with warm-up and per-track cooldown logic to prevent duplicate events.

Tools: Python, OpenCV, Ultralytics YOLO, NumPy, JSON

GitSetLive – AI-Powered GitHub Repository Management Platform

[GitSetLive](#)

- Built intelligent repo-management platform with GitHub OAuth, automated auditing, and AI-assisted analysis workflows.
- Implemented self-healing (Auto-PR) AI agent using Groq API for automated project bug correction, saving user time.

Tools: Python, FastAPI, GitHub API, Cloudflared, GitHub OAuth, SMTP, Groq API

NEmonsoonAI – ML Waterborne Disease Prediction and Safety System

[NEmonsoonAI](#)

- Processed health datasets with Pandas/NumPy; engineered features and trained Scikit-Learn classifiers for early-risk detection.
- Designed analytics dashboards to support data-driven community health decision-making.

Tools: Python, Pandas, NumPy, Scikit-Learn, Streamlit, Open-Meteo API, Google News RSS, Groq API

EXPERIENCE

Vice President – GRID Community

2025 – Present

- Led operations for a 2000+ member technical community; organised 6 industry-speaker events with end-to-end logistics.
- Drove community growth through structured engagement programmes and outreach initiatives.

Technical Member – ISTE-HIT-SC

2025 – Present

- Served as a Technical Member of the institution's ISTE chapter, supporting technical initiatives and events.
- Volunteered and coordinated 10+ technical events, handling planning, logistics, and on-ground execution.

Project Admin (LACOPRED) – ISTE Open Source 101

2025 | Remote

- Coordinated contributor onboarding, issue tracking, and documentation workflows across the open-source programme.
- Facilitated collaboration between contributors and maintainers through GitHub pull requests and code reviews.

Open Source Contributor – Resourcio Aperture 3.0 & ELUSOC

2025 | Remote

- Enhanced project documentation and README workflows to improve contributor onboarding in both programmes.
- Applied Git/GitHub best practices for version control, pull requests, and distributed repository management.

SKILLS

Languages: Python, C++, Java, SQL, Bash, JavaScript

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Feature Engineering, Model Training, Model Evaluation, Deep Learning

Generative AI: LLMs, RAG, Vector Embeddings, LangChain, LangGraph, AI Agents

Computer Vision: OpenCV, YOLO (Ultralytics)

ML/AI Engineering: FastAPI, MLflow, Docker, Flask, CI/CD

Cloud & DevOps: AWS, Linux, Git, GitHub

Soft Skills: Leadership, Team Management, Public Speaking, Community Building

ACHIEVEMENTS

- 1st Position – InnovateX Hackathon 2025
- 3rd Position – National Level Ideathon 2025
- Participant – Smart India Hackathon (SIH) 2025 and 2026
- Project Admin – ISTE Open Source 101 | Contributor – Resourcio Aperture 3.0 & ELUSOC

CONFERENCES & EVENTS

Bitcoin India Conference 2025 (Attendee) | HTF Buildathon Kolkata 2025 (Attendee) | MLH Diversion 2026 (Top 20) | MLH Hacknex S2 (Top 5)